Deep Learning for Dimensionality Reduction in Multi-Excitation Hyperspectral Imaging

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***Abstract*—Conventional Hyperspectral Imaging (HSI) is based on acquiring either reflectance, emission, or excitation profiles from each pixel of an image. Emission-based HSI can then be expanded into a fourth dimension by varying the excitation wavelength, which enables an improved discrimination of materials that contain mixed fluorophores. Multi-excitation HSI (ME-HSI) creates a 4D dataset encompassing signal intensity across two spatial coordinates, emission and excitation wavelengths. The processing of ME-HSI datasets present notable challenges due to the numerous spectral layers per pixel, introducing considerable redundancy and computational burden. Existing band selection methods, developed for a conventional 3D HSI, fail to capture the nonlinear cross-excitation correlations present in ME-HSI. To fill this gap, we developed a deep learning framework comprising three stages: (1) a 3D convolutional autoencoder with parallel excitation branches that learns a unified latent representation of ME-HSI data, (2) a perturbation-based attribution method that traces reconstruction sensitivity to individual excitation-emission wavelength pairs, and (3) Maximum Marginal Relevance (MMR) selection that balances informativeness with spectral diversity. Evaluated on lichen samples with four ground-truth classes across 3,072 configurations, the framework achieved 95.2% classification accuracy with only 80 selected bands (58% data reduction)—surpassing the 88.2% baseline obtained with all 192 bands. Even with 95% data reduction—which corresponds to only 9 out of 192 bands—accuracy exceeded baseline. The presented framework facilitates practical applications of ME-HSI by reducing acquisition complexity while improving discriminative power.**

***Index Terms*—Hyperspectral imaging, wavelength selection, convolutional autoencoder, perturbation analysis, spectral band selection, deep learning, fluorescence imaging**

I. Introduction

Hyperspectral imaging has emerged as a transformative technology for remote sensing, material characterization, and biomedical analysis, capturing reflected or emitted light across hundreds of contiguous wavelength bands—far exceeding the capabilities of conventional RGB or multispectral cameras [1], [2]. While reflectance-based HSI measures how materials scatter incident light, fluorescence-based HSI traditionally captures the emission spectra produced when materials absorb ultraviolet or visible light and re-emit at longer wavelengths [3]. This fluorescence response encodes molecular-level information about chromophores, enabling discrimination of materials that appear identical under reflectance imaging but possess distinct fluorescent signatures.

The multi-excitation HSI (ME-HSI) approach combines acquisition of emission profiles with stepwise changes in the wavelength of illuminating light, yielding a four-dimensional data structure spanning spatial coordinates (x, y), emission wavelength λ_{em}, and excitation wavelength λ_{ex}. This yields a simplified version of the excitation-emission matrix (EEM) for each spatial pixel—a comprehensive fluorescence fingerprint that has proven valuable for applications ranging from mineral identification [4] and vegetation stress monitoring [5] to biomedical tissue characterization [6] and cultural heritage analysis [7].

The discriminative power of ME-HSI comes at a computational cost. An example acquisition with $N_{\text{ex}} = 8$ excitation wavelengths and $N_{\text{em}} = 24$ emission bands per excitation yields 192 spectral layers per pixel, which for a megapixel image translates to nearly 200 million data points per frame. Not all of these excitation-emission combinations carry discriminative information: spectral overlap between fluorophores, detector noise in low-signal regions, and physical constraints such as Rayleigh scattering near the excitation wavelength introduce redundancy. The fundamental question thus becomes: *which excitation-emission wavelength pairs actually matter for discriminating materials of interest?* Answering this question has direct implications for: (i) sensor design—reducing the number of required LEDs, filters, and detector channels; (ii) real-time processing—enabling faster classification; and (iii) system deployment—decreasing power consumption and data transmission requirements.

The challenge of identifying informative spectral bands has been extensively studied for conventional 3D HSI [8], [9], with approaches spanning filter-based methods (variance, entropy), wrapper methods (sequential forward selection, genetic algorithms), and embedded methods (sparse regression, attention mechanisms) [10], [11], [12], [13], [14], [15]. However, these approaches have notable limitations when applied to ME-HSI: filter-based methods lack task relevance, wrapper methods suffer from prohibitive computational cost when the search space is large, and embedded methods require labeled training data. Importantly, the vast majority of these methods treat each spectral band independently, assuming a 3D data structure of (x, y, λ) where bands can be selected without considering their relationship to excitation wavelengths. For ME-HSI data, where the information lies in specific excitation-emission *combinations* and nonlinear cross-excitation correlations may encode important discriminative patterns, methods that explicitly model the joint spectral-spatial-excitation structure are needed. To the authors' knowledge, no existing band selection method addresses this 4D structure directly.

Deep learning offers a principled approach to learning nonlinear representations from high-dimensional data without requiring explicit feature engineering [16]. Autoencoders, which learn compressed representations by training a network to reconstruct its inputs, provide an unsupervised mechanism for identifying the essential structure in data [17], [18]. An autoencoder trained on ME-HSI data must encode the information necessary to reconstruct all excitation-emission combinations; analyzing what the autoencoder has learned—specifically, probing the sensitivity of reconstruction to perturbations in the latent space—enables attribution of importance to individual wavelengths without requiring class labels. This insight connects representation learning to feature selection: the autoencoder identifies important features implicitly through compression, and perturbation analysis makes this implicit knowledge explicit.

This paper presents an unsupervised deep learning framework for wavelength selection in ME-HSI datasets, combining a 3D convolutional autoencoder with parallel excitation branches, perturbation-based attribution, and Maximum Marginal Relevance (MMR) selection [19]. The framework requires no class labels and produces ordered wavelength rankings at any desired reduction level; the methodology is detailed in Section II. To the authors' knowledge, this represents the first deep learning approach for wavelength selection in 4D ME-HSI data.

II. Methodology

A. Data Preprocessing

Raw ME-HSI acquisitions required several preprocessing steps before autoencoder training.

1) Intensity Normalization: Fluorescence intensity depends on excitation source power and exposure time, which may vary across acquisitions. Since the LED-based multi-excitation source operated at different power levels and exposure times for each excitation wavelength, raw intensities were normalized to ensure comparability across excitation channels:

$$I_{\text{norm}}(x, y, \lambda_{\text{em}}, \lambda_{\text{ex}}) = \frac{I_{\text{raw}}(x, y, \lambda_{\text{em}}, \lambda_{\text{ex}})}{t_{\text{exp}}} \times \frac{t_{\text{ref}} \cdot P_{\text{ref}}}{P(\lambda_{\text{ex}})} \quad (1)$$

where t_{exp} is the exposure time and $P(\lambda_{\text{ex}})$ is the lamp power at excitation wavelength λ_{ex}. The reference values t_{ref} and P_{ref} were set to the maximum exposure time and maximum lamp power, respectively, across all excitation wavelengths, ensuring that the normalization factor scales intensities upward for weaker or shorter-exposed channels. Global min-max normalization was then applied to scale all values to $[0, 1]$.

2) Rayleigh Scattering Removal: Elastic scattering produces artifacts at two spectral locations. First-order Rayleigh scattering occurs when emission wavelength approaches excitation wavelength ($\lambda_{\text{em}} \approx \lambda_{\text{ex}}$); a spectral cutoff with offset $\Delta\lambda = 40\,\text{nm}$ excluded emission bands where $\lambda_{\text{em}} < \lambda_{\text{ex}} + \Delta\lambda$. Second-order Rayleigh scattering produces artifacts near $\lambda_{\text{em}} \approx 2\lambda_{\text{ex}}$; emission bands within $\pm 40\,\text{nm}$ of this harmonic were also excluded. Together, these cutoffs resulted in variable numbers of valid emission bands per excitation, yielding 192 valid excitation-emission pairs from the original acquisition grid.

3) Background Masking: Non-sample regions (background) provide no discriminative information and may bias the autoencoder toward reconstructing uninformative patterns. Binary masks $M(x, y) \in \{0, 1\}$ were generated through intensity thresholding or manual annotation, excluding background pixels from both training and evaluation.

B. 3D Convolutional Autoencoder Architecture

The central architectural challenge in processing 4D ME-HSI data was the variable spectral dimension across excitations. After Rayleigh cutoff, each excitation wavelength λ_{ex} yielded a different number of valid emission bands $N_{\text{em}}(\lambda_{\text{ex}})$. A monolithic architecture would require padding and masking complexities; instead, a parallel-branch design was employed where each excitation is processed by a dedicated encoder branch, with features merged into a shared latent representation.

1) Encoder Architecture: For each excitation wavelength $\lambda_{\text{ex}}^{(i)}$, $i = 1, \ldots, N_{\text{ex}}$, an encoder branch was defined, comprising a 3D convolutional layer that processes the spatial-spectral cube:

$$\mathbf{h}^{(i)} = \sigma\left(\text{Conv3D}\left(\mathbf{X}^{(i)}\right)\right) \quad (2)$$

where $\mathbf{X}^{(i)} \in \mathbb{R}^{H \times W \times N_{\text{em}}^{(i)}}$ is the input cube for excitation i, Conv3D applies $k_1 = 20$ filters of size $5 \times 5 \times \min(5, N_{\text{em}}^{(i)})$, and σ is the sigmoid activation function. The adaptive spectral kernel size accommodated variable band counts across excitations.

After convolution, adaptive average pooling standardized the spectral dimension to 1, yielding feature maps $\mathbf{h}^{(i)} \in \mathbb{R}^{k_1 \times H \times W}$ with consistent dimensions across excitations. Dropout regularization (rate 0.5) was applied after pooling to prevent overfitting.

2) Feature Merging: The excitation-specific feature maps were combined through element-wise averaging:

$$\bar{\mathbf{h}} = \frac{1}{N_{\text{ex}}} \sum_{i=1}^{N_{\text{ex}}} \mathbf{h}^{(i)} \quad (3)$$

This merging strategy offers several advantages over alternatives. Concatenation would yield feature dimensionality that grows with N_{ex}, increasing model complexity. Averaging provided implicit noise reduction and was robust to intensity variations across excitations. The merged features $\bar{\mathbf{h}}$ captured cross-excitation correlations by integrating patterns from all excitation branches.

3) Shared Latent Space: An additional convolutional layer transformed the merged features into the latent representation:

$$\mathbf{z} = \sigma\left(\text{Conv3D}\left(\bar{\mathbf{h}}\right)\right) \quad (4)$$

where $\mathbf{z} \in \mathbb{R}^{k_3 \times 1 \times H \times W}$ with $k_3 = 20$ channels constitutes the bottleneck encoding. This latent space $\mathbf{z}$ encapsulates the essential structure of the 4D ME-HSI data in a compressed form. The perturbation analysis (Section II-C) probes this space to identify which wavelengths contribute most to the learned representation.

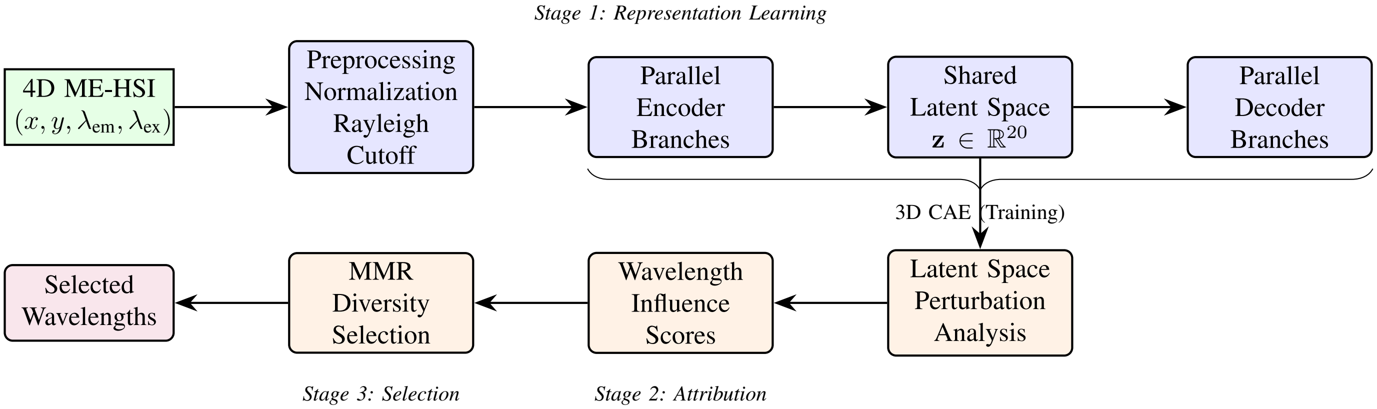


Fig. 1: Overview of the proposed wavelength selection framework. The pipeline comprises three stages: (1) a 3D convolutional autoencoder learns a compressed representation of 4D ME-HSI data; (2) perturbation analysis attributes latent dimensions to individual wavelengths; (3) MMR selection identifies diverse, informative band subsets.

4) Decoder Architecture: The decoder mirrors the encoder with symmetric structure. A shared decoding layer transformed the latent representation:

$$\mathbf{g} = \sigma\left(\text{Conv3D}\left(\mathbf{z}\right)\right) \tag{5}$$

yielding features $\mathbf{g} \in \mathbb{R}^{k_1 \times 1 \times H \times W}$. Parallel decoder branches then reconstructed each excitation's spectral cube:

$$\hat{\mathbf{X}}^{(i)} = \sigma\left(\text{Conv3D}^{(i)}\left(\mathbf{g}\right)\right) \tag{6}$$

where $\text{Conv3D}^{(i)}$ maps from k_1 to $N_{\text{em}}^{(i)}$ channels. Figure 2 illustrates the complete architecture.

5) Loss Function and Training: The autoencoder was trained to minimize masked mean squared error:

$$\mathcal{L} = \frac{1}{N_{\text{valid}}} \sum_{i=1}^{N_{\text{ex}}} \sum_{x,y} M(x, y) \cdot \left\| \mathbf{X}_{x,y}^{(i)} - \hat{\mathbf{X}}_{x,y}^{(i)} \right\|^2 \tag{7}$$

where $M(x, y)$ is the background mask, N_{valid} is the number of valid pixels, and the sum runs over all excitations and spatial locations. Masking ensured the network focused on sample regions rather than background.

Training employed the Adam optimizer [20] with learning rate $\eta = 0.001$, learning rate scheduling (ReduceOnPlateau with factor 0.5 and patience 5), and early stopping (patience 5 epochs). Large images were partitioned into 64×64 patches with 8-pixel overlap for batch training; typical convergence occurred within 25–30 epochs.

C. Perturbation-Based Wavelength Attribution

The trained autoencoder learned a compressed representation that captures the essential structure of ME-HSI data. Different latent dimensions encode different aspects of this structure; the goal was to attribute each latent dimension's influence back to specific excitation-emission wavelength pairs. If perturbing a latent dimension significantly changed the reconstruction of a particular wavelength, that wavelength must be encoded by that dimension.

1) Latent Dimension Scoring: Not all latent dimensions carry equal information. The $k_3 = 20$ latent dimensions were first ranked by importance using variance-based scoring:

$$\text{score}(d) = \text{Var}_{\text{samples}}\left(\mathbf{z}[\cdot, d, \cdot, \cdot, \cdot]\right) \tag{8}$$

where $\mathbf{z}[\cdot, d, \cdot, \cdot, \cdot]$ denotes dimension d across all samples and spatial locations. Higher variance indicated that the dimension captured meaningful variation rather than constant or near-zero values. The top k dimensions were then selected for perturbation analysis, where k is treated as a hyperparameter; empirical evaluation (Section IV) revealed that $k = 1$ (using only the highest-variance dimension) achieved optimal performance.

Alternative scoring methods include activation magnitude (mean absolute value) and reconstruction sensitivity (MSE when dimension is zeroed). Variance-based scoring yielded the most consistent wavelength rankings, likely because high-variance dimensions captured the primary modes of spectral variation that distinguish different materials.

2) Systematic Perturbation: For each selected latent dimension d, the standard deviation σ_d was computed across spatial locations and samples, and then the perturbed representations were generated:

$$\mathbf{z}_d^+ = \mathbf{z} + \epsilon \cdot \sigma_d \cdot \mathbf{e}_d \tag{9}$$

$$\mathbf{z}_d^- = \mathbf{z} - \epsilon \cdot \sigma_d \cdot \mathbf{e}_d \tag{10}$$

where $\mathbf{e}_d$ is a one-hot vector selecting dimension d, and ϵ controls perturbation magnitude. Values of $\epsilon \in \{15, 30, 45\}$ were used, and results across scales were aggregated to ensure robustness. The σ-based scaling ensured perturbations were proportional to the natural variation in each dimension, avoiding both negligibly small and destructively large perturbations.

3) Sensitivity Mapping: Decoding the perturbed representations yielded reconstructions $\hat{\mathbf{X}}_d^+$ and $\hat{\mathbf{X}}_d^-$. The sensitivity of each wavelength to perturbation in dimension d was computed:

$$\Delta_d(\lambda_{\text{ex}}, \lambda_{\text{em}}) = \frac{\left|\hat{X}_d^+(\lambda_{\text{ex}}, \lambda_{\text{em}}) - \hat{X}_d^-(\lambda_{\text{ex}}, \lambda_{\text{em}})\right|}{2\epsilon \cdot \sigma_d} \tag{11}$$

where averaging was performed over spatial locations and samples. This finite difference approximation estimates the

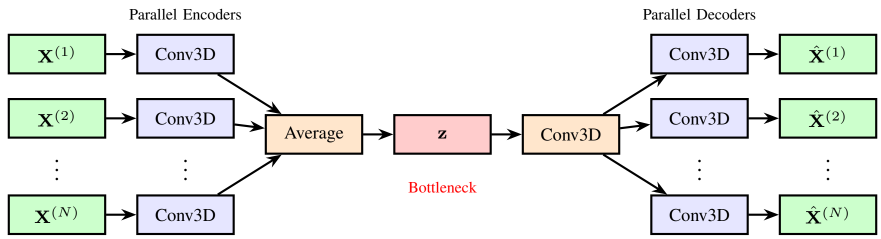


Fig. 2: 3D CAE architecture with parallel branches. Each excitation is processed by a dedicated encoder branch; features are averaged into a shared latent space; parallel decoder branches reconstruct the original 4D structure.

partial derivative of reconstruction with respect to latent dimension d, quantifying how strongly each wavelength's reconstruction responds to changes in that dimension.

4) Influence Accumulation: Sensitivity was accumulated across latent dimensions, weighted by dimension importance:

$$\text{Influence}(\lambda_{\text{ex}}, \lambda_{\text{em}}) = \sum_{d \in \text{top-}k} \text{score}(d) \cdot \overline{\Delta_d(\lambda_{\text{ex}}, \lambda_{\text{em}})} \quad (12)$$

where $\overline{\Delta_d}$ denotes the mean sensitivity across perturbation magnitudes. The result is an influence matrix $\mathbf{I} \in \mathbb{R}^{N_{\text{ex}} \times \max(N_{\text{em}})}$ assigning importance scores to each excitation-emission pair.

D. Maximum Marginal Relevance Selection

Selecting the top-K wavelengths by influence score alone may yield redundant bands. Adjacent wavelengths often captured correlated fluorescence patterns; a diverse selection covering different spectral regions may be more informative than a concentrated selection. Maximum Marginal Relevance (MMR) [19] was adapted from information retrieval to balance relevance with diversity.

1) MMR Formulation: Given the set of all wavelength pairs $\mathcal{W}$ and a partial selection $\mathcal{S}$, MMR scores the next candidate as:

$$\text{MMR}(w_i) = \lambda \cdot \text{Rel}(w_i) - (1 - \lambda) \cdot \max_{w_j \in \mathcal{S}} \text{Sim}(w_i, w_j) \quad (13)$$

where $\text{Rel}(w_i)$ is the normalized influence score, $\text{Sim}(w_i, w_j)$ is the cosine similarity between spectral profiles at wavelengths w_i and w_j, and $\lambda \in [0, 1]$ controls the relevance-diversity trade-off.

At $\lambda = 1$, MMR reduces to pure relevance ranking; at $\lambda = 0$, selection maximizes diversity without regard for informativeness. Empirical evaluation (Section IV) identified $\lambda = 0.5$ as optimal, providing balanced consideration of both criteria.

2) Selection Procedure: Algorithm 1 presents the greedy MMR selection procedure. The algorithm iteratively selected the wavelength that maximized the MMR score until K wavelengths were chosen. The output is an ordered list of excitation-emission pairs, ranked by selection order, enabling flexible thresholding based on accuracy-efficiency trade-offs.

Algorithm 1 MMR Wavelength Selection

Require: Wavelength set $\mathcal{W}$, influence matrix $\mathbf{I}$, target count K, trade-off parameter λ
Ensure: Selected wavelength set $\mathcal{S}$

```
Initialize 𝒮 ← ∅
Compute Rel(w) ← I(w)/max(I) for all w ∈ 𝒲
for i = 1 to K do
    for each w ∈ 𝒲 \ 𝒮 do
        if 𝒮 = ∅ then
            MMR(w) ← Rel(w)
        else
            MMR(w) ← λ · Rel(w) − (1 − λ) · max_{w_j ∈ 𝒮} Sim(w, w_j)
        end if
    end for
    w* ← arg max_{w ∈ 𝒲\𝒮} MMR(w)
    𝒮 ← 𝒮 ∪ {w*}
end for
return 𝒮
```

E. Validation Protocol

The wavelength selection pipeline (Sections II-B–II-D) is entirely unsupervised and operates without access to class labels. The supervised classification described in this section serves *only* to evaluate the quality of selected wavelengths—it does not influence the selection process. In a deployment scenario, wavelength selection would proceed identically whether or not labeled data exists; the KNN classifier was applied post-hoc to quantify discriminative power. The parameter analysis in Section IV similarly represents retrospective characterization of method behavior, not optimization toward classification accuracy.

The selected wavelengths were validated through supervised classification using only the selected bands as features.

1) Feature Extraction: For each pixel (x, y) in regions with ground-truth labels, the spectral values were extracted at the selected wavelength pairs:

$$\mathbf{f}(x, y) = \left[I(x, y, \lambda_{\text{em}}^{(1)}, \lambda_{\text{ex}}^{(1)}), \ldots, I(x, y, \lambda_{\text{em}}^{(K)}, \lambda_{\text{ex}}^{(K)}) \right]^{\top} \quad (14)$$

where $(\lambda_{\text{em}}^{(k)}, \lambda_{\text{ex}}^{(k)})$ is the k-th selected pair. Features were standardized (zero mean, unit variance) before classification.

2) Classification: K-Nearest Neighbors (KNN) [21] with 5 neighbors and Euclidean distance was employed. KNN is non-parametric and makes no assumptions about data distribution, isolating the effect of wavelength selection from classifier capacity. This choice ensures that observed performance differences reflect the quality of selected wavelengths rather than

TABLE I: Lichen Dataset Specifications

Parameter	Value
Spatial dimensions	1040×925 pixels
Excitation wavelengths	8 (310, 325, 340, 365, 385, 400, 415, 430 nm)
Emission range	420–720 nm
Emission bands (per ex.)	22–28 (variable after Rayleigh cutoff)
Total EEM pairs	192
Spectral resolution	$\sim$12.5 nm
Ground-truth classes	4 lichen species

classifier-specific biases. Labeled regions of interest (ROIs) provided training and evaluation data.

3) Metrics: Classification accuracy, precision, recall, and F1-score (weighted by class frequency) were reported. Cohen's Kappa (κ) [22] quantifies agreement beyond chance, with $\kappa > 0.8$ indicating excellent agreement. For comparison, a baseline using the full 192-band spectrum with the same KNN classifier was established. Relative performance (selected accuracy divided by baseline accuracy) quantifies the accuracy-efficiency trade-off achieved by wavelength selection.

III. Experimental Setup

A. Datasets

The presented framework was evaluated on a multi-excitation hyperspectral fluorescence dataset of biological specimens.

1) Lichen Fluorescence Dataset: The primary dataset comprised ME-HSI acquisitions of lichen specimens—composite organisms formed by symbiotic associations between fungi and photosynthetic partners (algae or cyanobacteria). Lichens serve as important bioindicators for environmental monitoring, and their autofluorescence signatures reflect both fungal and algal components [23].

Table I summarizes the dataset details. The acquisition system employed an LED-based multi-excitation source (WheeLED, Mightex, Toronto, CA) scanning from 310–430 nm spanning the UV-A to visible range, paired with a Nuance FX hyperspectral camera (PerkinElmer, Waltham, MA) capturing emission spectra from 420 to 720 nm at each excitation. Exposure times and lamp power were recorded for intensity normalization per Equation (1).

Ground-truth annotations were generated based on the knowledge of the origins of each lichen obtained from the Takhtajyan Institute of Botany, National Academy of Sciences of Armenia. Four regions corresponding to different lichen species were delineated, providing ground-truth annotations for supervised evaluation. The total of 192 valid excitation-emission pairs (rather than $8 \times 29 = 232$ from the raw grid) results from variable emission windows after first- and second-order Rayleigh scattering removal (Section II-A).

B. Implementation Details

The framework was implemented in Python 3.10 using PyTorch 2.0 [24] for deep learning components and scikit-learn 1.3 [25] for classification and metrics. All experiments were conducted on a workstation with an NVIDIA RTX 4070 GPU (8 GB VRAM) and 64 GB system RAM.

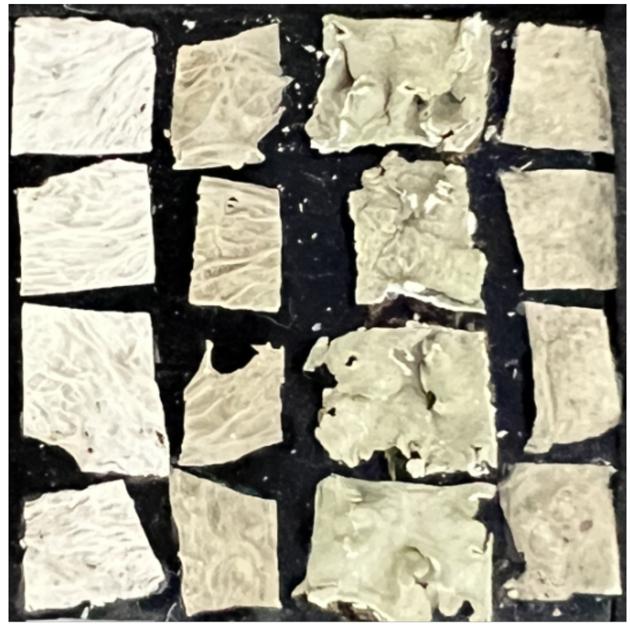


Fig. 3: Visual appearance of lichens used in this study. The first two columns are 7x7mm pieces of Ramalina Sinesis (back and front). The third and fourth columns are Flavopunctelia Soredica and Ramalina Fraxinea, respectively (both front surfaces).

TABLE II: Autoencoder Hyperparameters

Parameter	Value	Rationale
Encoder filters (k_1)	20	Capacity vs. complexity
Latent filters (k_3)	20	Bottleneck dimensionality
Spatial filter size	5×5	Local pattern capture
Spectral filter size	min(5, N_{em})	Variable band counts
Activation	Sigmoid	Match $[0, 1]$ input
Dropout rate	0.5	Regularization
Patch size	64×64	GPU memory constraints
Patch overlap	8 pixels	Boundary continuity
Batch size	32	Training stability
Optimizer	Adam [20]	Adaptive learning rates
Initial learning rate	0.001	Standard default
LR scheduler	ReduceOnPlateau	Convergence refinement
LR factor	0.5	Reduction magnitude
Patience (LR)	5 epochs	Schedule sensitivity
Early stopping patience	5 epochs	Prevent overfitting

1) Autoencoder Hyperparameters: Table II lists the autoencoder configuration. $k_1 = k_3 = 20$ filters were employed based on preliminary experiments showing this provided sufficient capacity without overfitting. The 5×5 spatial filter captured local texture patterns relevant for biological specimens. Sigmoid activations ensured outputs remained in $[0, 1]$, matching the normalized input range.

2) Perturbation Analysis Parameters: For latent space analysis, dimensions ranked by either variance or PCA contribution were explored at $k \in \{1, 3\}$, and perturbations were applied using two methods—percentile-based and absolute-range—at medium ($\epsilon \in \{30, 40, 50\}$) and high ($\epsilon \in \{50, 60, 70\}$) magnitude levels. Sensitivity values were averaged across 1,000 randomly sampled patches to ensure robust estimation. Additionally, three normalization strategies for influence

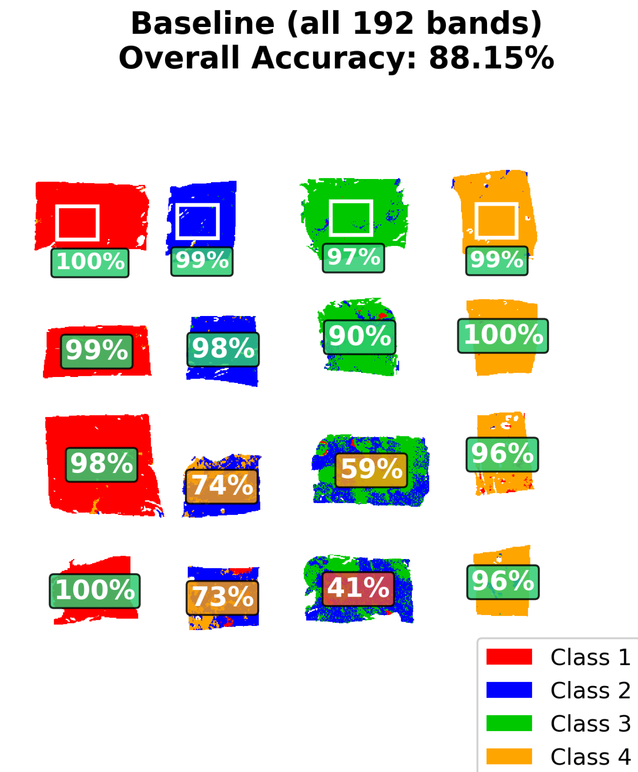


Fig. 4: Ground-truth annotation and baseline classification results using all 192 bands. The 16 lichen specimens are arranged in a 4×4 grid, with columns corresponding to the four classes (red = Class 1, blue = Class 2, green = Class 3, yellow = Class 4). Equal-sized square ROIs (one per class, approximately 50×60 pixels each) in the first row provide training exemplars for KNN evaluation, with per-region classification accuracies shown in callout boxes (11,519 total training pixels). These ROIs were used *only* for post-hoc validation of wavelength selection quality; they did not influence the unsupervised selection pipeline.

scores were compared: no normalization, max-per-excitation, and variance normalization (see Section IV).

3) MMR Configuration: The MMR selection used $\lambda = 0.5$ (equal weight to relevance and diversity), with spectral similarity computed as cosine similarity between full spectral profiles at each wavelength pair.

4) Classification Setup: K-Nearest Neighbors (KNN) classification employed $K = 5$ neighbors with Euclidean distance metric. Features were standardized using z-score normalization (zero mean, unit variance). For evaluation, all annotated pixels were used, with four regions of interest (ROIs) providing class exemplars (Figure 4); the same classifier configuration was applied consistently across all experiments for fair comparison.

C. Experimental Configurations

Multiple configurations were evaluated, varying the parameters of each framework component. Table III summarizes the configuration space explored.

TABLE III: Configuration Parameter Space

Parameter	Values Tested
Dimension selection method	Variance, PCA
Number of dimensions (k)	1, 3
Perturbation method	Percentile, Absolute range
Perturbation magnitudes (ϵ)	Medium $\{30{,}40{,}50\}$, High $\{50{,}60{,}70\}$
Normalization method	None, Max-per-excitation, Variance
MMR trade-off (λ)	0.5 (fixed)
Number of bands selected	1–180 (64 values)

From the full configuration space, a comprehensive sweep of 3,072 experiments was conducted spanning all parameter combinations:

- **Band count sweep:** 64 values from 1 to 180 bands (step 1 for 1–20, step 2 for 22–60, step 5 for 65–180).
- **Attribution parameters:** Two dimension selection methods $\times$ two dimension counts $\times$ two perturbation methods $\times$ two magnitude levels.
- **Normalization:** Three post-attribution normalization strategies to assess sensitivity to score scaling.

This exhaustive exploration characterized framework behavior across the hyperparameter space, identifying robust parameter combinations rather than optimizing for the classification task. Each configuration produced a wavelength ranking through the unsupervised pipeline; classification accuracy was measured post-hoc to assess quality. MMR diversity was fixed at $\lambda = 0.5$ based on prior analysis showing this provided the optimal relevance-diversity balance.

D. Evaluation Metrics

Wavelength selection was evaluated through multiple metrics capturing different aspects of performance.

1) Classification Metrics:

- **Accuracy:** Fraction of correctly classified pixels.
- **Balanced Accuracy:** Average of per-class recall, robust to class imbalance.
- **Precision/Recall/F1:** Weighted by class frequency to account for distribution differences.
- **Cohen's Kappa (κ):** Agreement beyond chance [22]; $\kappa > 0.8$ indicates excellent agreement.
- **Matthews Correlation Coefficient (MCC):** Balanced measure suitable for multi-class problems.

2) Reconstruction Metrics:

- **RMSE:** Root Mean Squared Error between input and reconstruction.
- **PSNR:** Peak Signal-to-Noise Ratio in decibels (dB).

3) Reduction Metrics:

- **Data Reduction:** $1 - K/N_{\text{total}}$, where K is the number of selected bands and $N_{\text{total}} = 192$.
- **Relative Performance:** Accuracy with selected bands divided by baseline accuracy, expressed as a percentage.

4) Baseline Comparison: The baseline used all 192 wavelength pairs with identical KNN classification, providing a reference for evaluating selection effectiveness. Relative performance and data reduction jointly quantify the trade-off between accuracy and efficiency.

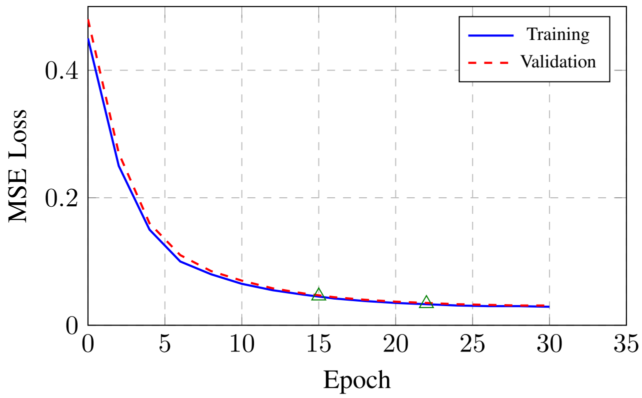


Fig. 5: Training convergence curves for the 3D CAE. Loss decreased smoothly from 0.45 to 0.03 over 30 epochs. Green triangles indicate learning rate reductions.

IV. Results

A. Autoencoder Performance

The 3D CAE converged reliably across multiple training runs, demonstrating stable representation learning for 4D ME-HSI data.

1) Training Convergence: Figure 5 shows representative training curves. The masked MSE loss decreased from approximately 0.45 to 0.03 over 25–30 epochs, with smooth convergence and no evidence of overfitting (training and validation losses tracked closely). Learning rate reductions triggered at epochs 15 and 22 refined convergence. Early stopping typically activated between epochs 25–30.

2) Reconstruction Quality: The trained autoencoder achieved RMSE of 0.078 and PSNR of 22.16 dB on the validation set. Visual inspection confirmed that reconstructions preserved key spectral features across all excitation wavelengths. Per-excitation analysis showed consistent reconstruction quality, with no systematic degradation at particular excitation wavelengths.

3) Latent Space Quality: Analysis of the 20-dimensional latent space revealed meaningful variation. The top 5 dimensions by variance accounted for 72% of total latent variance, indicating that information was distributed but concentrated in a subset of dimensions. This supported using top-k dimension selection for perturbation analysis.

B. Baseline Classification Performance

Using all 192 wavelength pairs with KNN classification established the performance ceiling. Table IV presents baseline metrics.

The baseline achieved 88.2% accuracy and $\kappa = 0.842$ [22], indicating excellent agreement with ground truth. Per-class analysis revealed performance varied across lichen classes:

- **Class 1:** 99.1% recall, 98.0% precision (best performing)
- **Class 2:** 86.6% recall, 66.8% precision (precision-limited)
- **Class 3:** 71.1% recall, 99.4% precision (recall-limited)
- **Class 4:** 97.8% recall, 88.6% precision (balanced)

TABLE IV: Baseline Performance (192 Bands)

Metric	**Value**
Accuracy	0.8815
Balanced Accuracy	0.8867
Precision (weighted)	0.9031
Recall (weighted)	0.8815
F1-Score (weighted)	0.8824
Cohen's Kappa (κ)	0.8416
Matthews Correlation	0.8478

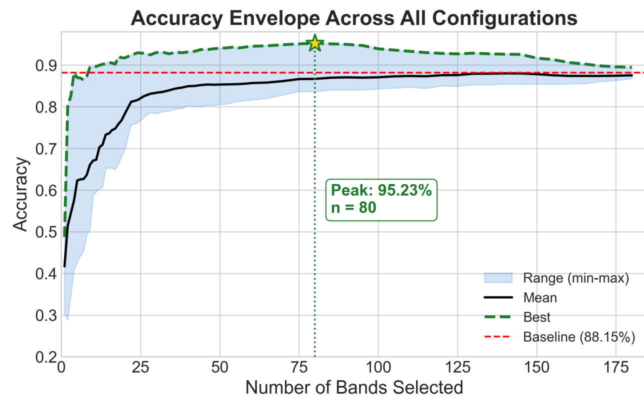


Fig. 6: Classification accuracy envelope as a function of band count. Shaded region: min–max range; solid line: mean. Peak accuracy (95.2%) occurs at 80 bands, surpassing the 88.2% baseline (red dashed).

Classes 2 and 3 showed confusion with each other (15,846 Class 3 pixels misclassified as Class 2), suggesting spectral overlap in their fluorescence signatures. This confusion pattern persisted across wavelength selections, indicating a fundamental limitation of the spectral data rather than the selection method.

C. Wavelength Selection Results

Wavelength selection was evaluated across 3,072 configurations spanning the full parameter space (Table III). Table V presents the best-performing configurations at key band counts, all of which used PCA-based dimension selection (see Section IV-F for method comparison).

1) Key Finding: Performance Exceeds Baseline: An interesting finding is that carefully selected subsets *exceeded* baseline performance. With 80 bands (58.3% data reduction), the method achieved **95.2% accuracy**—7.1 percentage points above the 88.2% baseline. Even with only 9 bands (95.3% data reduction), accuracy remained at 89.4%, still exceeding baseline. This demonstrates that the full spectrum contains redundant or noisy bands that hinder classification, and that peak performance occurs at an intermediate selection level. Performance remained above 90% across a wide range (13–100 bands), indicating robust selection; even the minimal 3-band configuration achieved 82.7%—far above random chance (25%).

Figure 6 visualizes the accuracy-reduction trade-off across all 3,072 configurations.

TABLE V: Wavelength Selection Performance Across Band Counts (Best Configuration per Band Count)

Configuration	Bands	Reduction	Accuracy	F1 (wtd)	κ	Rel. Perf.
Baseline (all 192 bands)	192	0.0%	0.8815	0.8824	0.8416	100.0%
PCA, 3-dim, 80 bands	**80**	**58.3%**	**0.9523**	**0.9523**	**0.9357**	**108.0%**
PCA, 1-dim, 60 bands	60	68.8%	0.9451	0.9451	0.9260	107.2%
PCA, 3-dim, 50 bands	50	74.0%	0.9404	0.9407	0.9198	106.7%
PCA, 1-dim, 40 bands	40	79.2%	0.9315	0.9318	0.9077	105.7%
PCA, 1-dim, 30 bands	30	84.4%	0.9300	0.9305	0.9058	105.5%
PCA, 1-dim, 20 bands	20	89.6%	0.9171	0.9181	0.8887	104.0%
PCA, 3-dim, 13 bands	13	93.2%	0.9016	0.9024	0.8684	102.3%
PCA, 3-dim, 10 bands	10	94.8%	0.8974	0.8980	0.8629	101.8%
PCA, 3-dim, 9 bands	**9**	**95.3%**	**0.8943**	**0.8948**	**0.8588**	**101.5%**
PCA, 1-dim, 5 bands	5	97.4%	0.8698	0.8713	0.8260	98.7%
PCA, 1-dim, 3 bands	3	98.4%	0.8267	0.8274	0.7684	93.8%

Note: Best configuration at each band count shown; all use PCA-based dimension selection with MMR $\lambda = 0.5$. Bold rows indicate key operating points. Rel. Perf. = accuracy/baseline accuracy.

TABLE VI: Robustness Analysis: Learned vs. Random Selection (13 Bands)

Metric	Value
Learned Selection	**0.9016**
Random Mean	0.4606
Random Std	0.0358
Random Minimum	0.3797
Random Maximum	0.5792
Random Median	0.4494
Learned vs. Random	**Outperformed all 10,000**

2) Visual Classification Results: Figure 7 presents spatial classification maps for the baseline, 80-band, and 9-band configurations. The 80-band selection (b) shows markedly improved spatial homogeneity compared to the baseline (a), with reduced Class 2/Class 3 confusion and sharper specimen boundaries. The 9-band selection (c) remains visually comparable to the baseline despite 95.3% data reduction. Residual confusion between Class 2 and Class 3 persists across all configurations, reflecting genuine spectral overlap between these Foliose morphological types rather than selection failure.

D. Robustness Validation

To verify that the method's performance reflected genuine wavelength selection intelligence rather than chance, a comprehensive robustness analysis was conducted.

1) Random Selection Comparison: The 13-band selection was compared against 10,000 randomly generated 13-band combinations from the 192 available wavelengths. Table VI and Figure 8 present the results.

The results demonstrated lear advantage of the learned selection (Figure 8):

- The method achieved 90.2% accuracy compared to a random mean of 46.1%—an improvement of 44 percentage points.
- The best random selection (57.9%) fell 32 percentage points below the learned result.
- The selection **surpassed every random combination**: it outperformed all 10,000 random samples tested.

TABLE VII: Object-Level Performance Summary

Statistic	Baseline	Best Selection
Mean Accuracy	0.886	0.947
Std Deviation	0.172	0.079
Minimum	0.409 (Obj. 12)	0.689 (Obj. 8)
Maximum	0.998 (Obj. 4)	1.000 (Obj. 1)
Objects Improved	–	15/16 (94%)
Objects Maintained	–	0/16 (0%)
Objects Declined	–	1/16 (6%)

This validated that the perturbation-based attribution successfully identified genuinely informative wavelengths rather than benefiting from fortuitous combinations.

2) Statistical Significance: The learned selection accuracy (0.9016) lay 12.3 standard deviations above the random mean:

$$z = \frac{0.9016 - 0.4606}{0.0358} = 12.3\sigma \tag{15}$$

This indicated statistical significance far beyond any reasonable threshold ($p \ll 10^{-20}$).

E. Object-Level Analysis

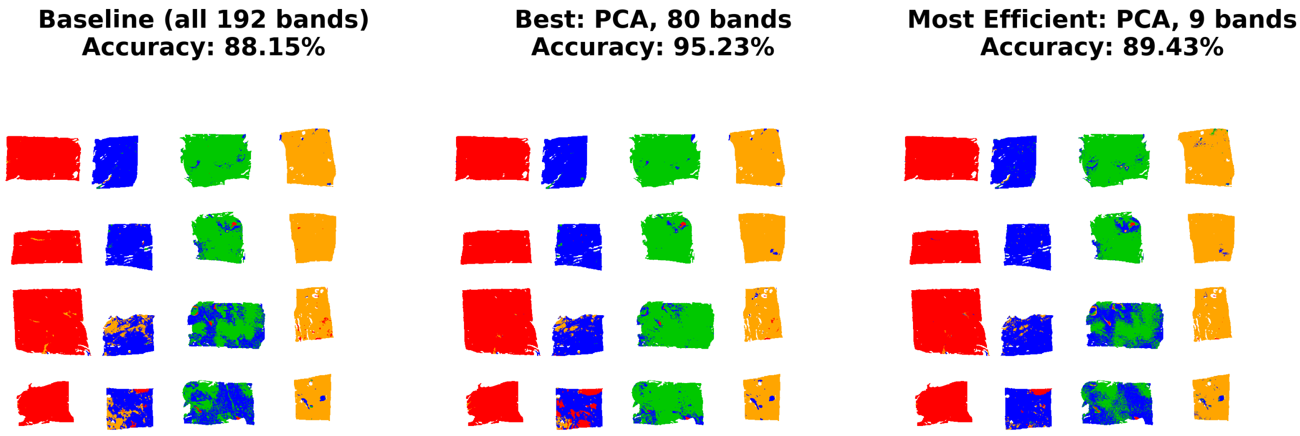
Object-level analysis was conducted across 16 individual lichen specimens to understand how wavelength selection affected different samples.

1) Performance Distribution: Table VII summarizes object-level performance statistics.

2) Challenging Objects Benefit Most: An interesting finding was that difficult-to-classify objects showed the most improvement:

- **Object 12** (Class 3, lowest baseline): Accuracy improved from 40.9% to 86.0% (+110.3% relative improvement).
- **Object 11** (Class 3): Improved from 58.5% to 91.7% (+56.8% relative improvement).
- **Object 8** (Class 2): Improved from 72.8% to 68.9%—the only object showing slight decline, reflecting the persistent Class 2/3 spectral overlap.
- **Object 7** (Class 2): Improved from 74.0% to 88.0% (+18.9% relative improvement).

This pattern suggested that the full 192-band spectrum contained wavelengths that were *detrimental* to classification of



(a) Baseline: 192 bands 88.2% accuracy

(b) 80 bands (58% reduction) 95.2% accuracy

(c) 9 bands (95% reduction) 89.4% accuracy

Fig. 7: Spatial classification maps. Colors indicate predicted class: red = Class 1, blue = Class 2, green = Class 3, yellow = Class 4. The 16 specimens are arranged in a 4×4 grid matching Figure 3.

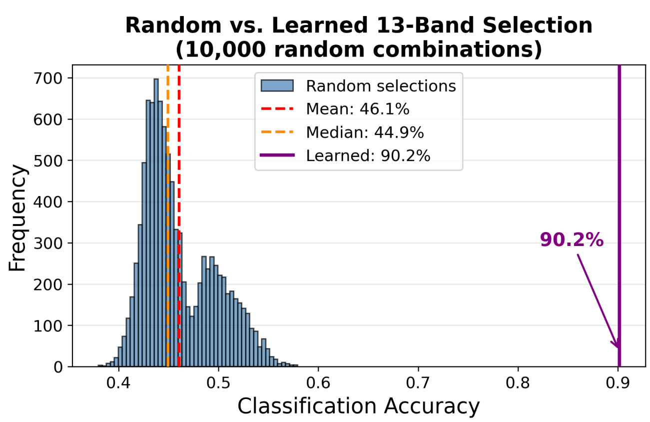


Fig. 8: Distribution of classification accuracies for 10,000 randomly sampled 13-band combinations. The histogram shows that random selections cluster around 46% accuracy (dashed lines indicate mean and median), while the learned selection (solid purple line at 90.2%) lies far in the upper tail, outperforming every random combination tested.

certain specimens—likely due to noise or interference patterns specific to their spectral signatures. Wavelength selection removed these harmful bands, with 15 of 16 specimens showing improvement and performance standard deviation decreasing from 0.172 to 0.079.

F. Configuration Parameter Analysis

The effect of key configuration parameters on selection performance was analyzed.

1) Dimension Selection Method: Table VIII compares dimension selection strategies. Best accuracy at any band count is shown for each method to characterize the performance ceiling of each approach.

TABLE VIII: Comparison of Dimension Selection Methods

Method	Best Acc.	κ	At n bands
PCA (latent)	**0.9523**	**0.9357**	**80**
Variance (latent)	0.8695	0.8257	175

PCA-based latent dimension selection outperformed variance-based selection (95.2% vs. 87.0% best accuracy). This finding demonstrated that PCA decomposition of the learned latent space more effectively identified the dimensions encoding discriminative information. Notably, PCA achieved its peak accuracy at 80 bands, while variance-based selection required 175 bands to reach its (lower) peak—suggesting that PCA selects more informative wavelengths early in the ranking. The variance-based method never exceeded the 88.2% baseline at any band count, while every PCA configuration from 9 bands upward surpassed it.

2) MMR Trade-off Parameter: Based on preliminary analysis showing $\lambda = 0.5$ optimally balanced relevance and diversity, the comprehensive sweep fixed this parameter. Lower values of λ prioritize diversity, potentially selecting less informative but spectrally distinct bands, while higher values prioritize relevance, leading to redundant selections concentrated in spectrally adjacent regions. The effectiveness of this balance is demonstrated in the wavelength analysis below (Section IV-G).

G. Selected Wavelength Analysis

Figure 9 presents a heatmap of mean wavelength importance across the 16 above-baseline configuration groups. For each group, the complete wavelength ranking from the maximum-band experiment was used, and a normalized importance score was computed as $1 - (\text{rank} - 1)/\text{max\_rank}$, then averaged across groups. The heatmap reveals that importance is not uniformly distributed across the EEM space. Emission wavelengths in the 480–620 nm range show the highest mean importance, corresponding to key fluorophore emission regions in biological specimens. The mid-range excitation wavelengths (325–400 nm) contribute the most consistently important bands, while the extreme excitation wavelengths (310 nm and 430 nm) show more concentrated emission preferences. Grey cells indicate wavelengths excluded by Rayleigh scattering cutoffs.

The 9-band optimal configuration exemplifies the framework's selection strategy: it selects exactly one wavelength from each of the 8 excitation sources (310–430 nm), with emission peaks spanning 500–680 nm. This one-per-excitation pattern suggests that the framework identifies the single most informative emission band at each excitation, consistent with the parallel-branch architecture that processes each excitation independently before feature averaging.

V. Discussion

A. Why Does the Method Exceed Baseline Performance?

The most counterintuitive result is that wavelength selection *exceeded* baseline performance (Table V). This finding—that fewer bands yield better classification—has important theoretical and practical implications.

Accuracy did not decrease monotonically with band reduction; instead, an optimal operating point existed where sufficient discriminative information was retained while noise was maximally suppressed. Beyond this optimum, further reduction traded accuracy for efficiency—a favorable exchange for resource-constrained applications.

1) Noise Removal Through Intelligent Selection: Hyperspectral data inherently contains redundancy and noise. Spectral bands with low signal-to-noise ratios, bands capturing uninformative variance (e.g., illumination artifacts), and bands with high inter-class overlap all contribute to the full spectrum. A classifier operating on all 192 bands had to navigate this noise, which could degrade decision boundaries.

By selecting only wavelengths with high discriminative value, the method effectively performed dimensionality reduction that *removed* uninformative or harmful information. The perturbation analysis identified wavelengths that the autoencoder considered important for faithful reconstruction—wavelengths that captured meaningful spectral structure rather than noise. This filtering mechanism explained why fewer bands could yield better classification.

2) Joint 4D Representation: Traditional band selection methods process each spectral band independently, assuming that informativeness can be assessed from individual band statistics. For ME-HSI data, where discriminative patterns emerge from excitation-emission *combinations*, this assumption failed. The parallel-branch architecture with feature averaging enabled the network to learn cross-excitation correlations—patterns that manifested only when comparing fluorescence responses across excitation wavelengths. The latent space encoded these joint patterns, capturing information that per-band analysis would miss.

3) Perturbation as Learned Attribution: The perturbation analysis operated on what the autoencoder had learned, not on raw data statistics. By measuring reconstruction sensitivity to latent perturbations, wavelengths that the network considered essential for accurate reconstruction were identified. This is fundamentally different from variance-based band selection, which might prioritize noisy bands with high variance but low discriminative value. The autoencoder's compression objective acted as an implicit importance filter: wavelengths that did not contribute to reconstruction fidelity were effectively ignored.

B. Robustness Analysis Insights

The robustness analysis (Section IV-D) provided strong evidence that the method exhibited genuine "intelligence" in wavelength selection.

1) Significance of the Random Comparison: The search space of $\binom{192}{13} \approx 3.6 \times 10^{18}$ possible 13-band combinations is far too large for exhaustive search. That the learned selection outperformed all 10,000 random samples—with a 12.3σ separation from the random mean ($p \ll 10^{-20}$)—demonstrated that the perturbation-based attribution genuinely identified discriminative wavelengths rather than benefiting from chance.

C. Object-Level Performance Patterns

The object-level analysis revealed important patterns with implications for practical deployment.

1) Improvement for Challenging Specimens: The object-level analysis (Table VII) revealed that difficult-to-classify specimens benefited most from wavelength selection. This suggests that the full spectrum contains wavelengths *detrimental* to certain specimens—likely bands where intra-class variance is high or spectral signatures overlap with other classes. Removing these bands preferentially improves challenging cases while maintaining performance on straightforward ones. This property is clinically valuable: in biomedical imaging, challenging specimens are often the most diagnostically important (e.g., early-stage lesions with subtle spectral signatures).

The accompanying 54% reduction in performance variance across specimens (Table VII) makes classification more predictable—important for quality control applications.

D. PCA vs. Variance Dimension Selection

The substantial gap between PCA and variance-based selection (Table VIII) reflects a fundamental difference in how these methods analyze the latent space. PCA identifies orthogonal directions in the autoencoder's *learned* representation, capturing structured variation through linear combination of dimensions. Variance-based selection ranks individual dimensions independently, missing these interactions—hence its lower performance ceiling and need for more bands to reach it.

E. The Role of Diversity

Pure relevance-based selection ($\lambda = 1$ in MMR) underperformed balanced selection ($\lambda = 0.5$). This counterintuitive

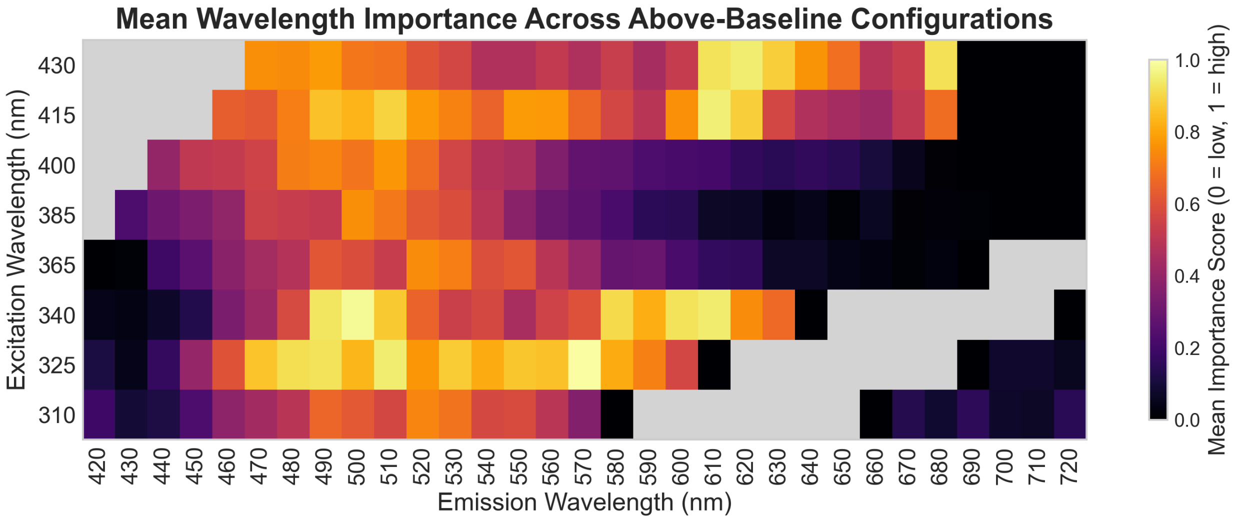


Fig. 9: Mean wavelength importance score across the 16 above-baseline configuration groups. Each cell shows the average normalized rank (1.0 = consistently top-ranked, 0.0 = lowest-ranked) for each excitation-emission pair. Grey cells indicate wavelengths excluded by Rayleigh scattering cutoffs.

result reflected the redundancy structure of fluorescence spectra.

High-influence wavelengths often clustered spectrally: if one emission wavelength at 480 nm was highly influential, adjacent wavelengths at 470 nm and 490 nm likely showed similar influence due to spectral overlap. Selecting all three provided little additional information but occupied three of the K available selection positions. MMR's diversity term penalized spectrally similar selections, encouraging the algorithm to "spread out" across the EEM space.

The optimal $\lambda = 0.5$ balanced these considerations: wavelengths had to be both relevant (high influence) and complementary (spectrally distinct). This mirrored findings in information retrieval, where query-relevant documents are more useful when they also provide novel information [19].

F. Limitations

Several limitations of the current study are acknowledged.

1) Dataset Scope: Evaluation focused on a single primary dataset (lichen fluorescence) with four classes. While results were encouraging, validation across diverse specimen types, imaging conditions, and larger class vocabularies would strengthen generalizability claims. Future work should evaluate the framework on datasets from different application domains, such as biomedical tissue imaging or agricultural produce inspection.

2) Ground Truth Quality: The four-class ground truth was generated through expert annotation, which may not perfectly capture biological reality. Some lichen regions may contain mixed species or transitional zones. The confusion between Class 2 and Class 3 (both Foliose morphological types; 15,846 misclassified pixels at baseline) may reflect genuine spectral similarity arising from their similar growth structures rather than method failure.

3) Computational Cost: While wavelength selection enabled efficient inference, the training pipeline—autoencoder training plus perturbation analysis—required computational power. For the lichen dataset, the complete pipeline took approximately 2 hours on the hardware used (30 minutes for autoencoder training, 90 minutes for perturbation analysis across multiple configurations). For rapid exploratory analysis, this overhead may be prohibitive, though it remained a one-time cost per dataset.

4) Black-Box Attribution: Perturbation analysis revealed *which* wavelengths the network considered important but not *why*. Physical interpretation (e.g., attributing high-influence wavelengths to specific fluorophores such as chlorophyll or lignin) requires domain expertise and cannot be derived from the method alone. Incorporating spectroscopic knowledge bases could enable more interpretable wavelength attributions.

5) Threshold Selection: The number of bands to select (K) is user-specified. While the results showed that 9 bands exceeded baseline performance for this dataset, choosing an appropriate K for a different application requires domain knowledge or cross-validation. Automated threshold selection based on performance curves remains an open problem.

G. Practical Implications

The framework offered practical value for ME-HSI system design and deployment.

1) Sensor Design Guidelines: The one-per-excitation pattern observed in the 9-band configuration (Section IV-G) directly informs sensor design: coverage across the full excitation range is essential, but only a single emission band per excitation is needed for effective discrimination. The 95.3% data reduction translates to proportional reductions in sensor complexity, cooling requirements, and power consumption.

2) Real-Time Processing and Data Transmission: Processing 9 features per pixel is 21.3× faster than 192 features for

KNN classification, with proportional memory savings. For embedded systems, UAV-mounted sensors, or clinical point-of-care devices, this reduction enables practical deployment. In remote sensing with bandwidth constraints, selective acquisition of only informative wavelengths could increase effective observation rates.

3) Training Data Efficiency: Because wavelength selection is unsupervised (the autoencoder requires no labels), the framework could identify informative wavelengths from unlabeled data. Subsequent supervised training for classification then operates on the reduced feature set, potentially requiring fewer labeled examples for equivalent performance. This two-stage approach—unsupervised selection followed by supervised classification—is particularly valuable when labeled data is scarce or expensive to obtain.

H. Comparison to Alternative Approaches

Direct comparison to existing methods is complicated by the novelty of 4D ME-HSI analysis. Most band selection literature addresses 3D hyperspectral data from remote sensing; methods designed for EEM data focus on spectroscopic (non-imaging) applications.

For 3D hyperspectral band selection, recent deep learning approaches include attention-based networks [15] and reinforcement learning formulations [26]. These methods require labeled training data and optimize selection for specific classification tasks. The unsupervised approach traded task-specific optimization for generalizability: the same wavelength selection could serve multiple downstream tasks (classification, segmentation, anomaly detection) without retraining.

For EEM spectroscopy, parallel factor analysis (PARAFAC) [27] decomposes 3D EEM tensors into component spectra, enabling chemical identification. Extending PARAFAC to 4D imaging data with spatial dimensions is computationally challenging and loses spatial context that the convolutional approach preserved.

To the authors' knowledge, no prior work directly addressed wavelength selection for 4D ME-HSI data using deep learning. The framework thus provided a baseline for future research in this emerging area, with the potential for extension through attention mechanisms, variational formulations, or end-to-end optimization with task-specific losses.

VI. Conclusion

We have presented a deep learning framework for wavelength selection in multi-excitation hyperspectral imaging, addressing the challenge of identifying informative spectral layers in 4D fluorescence data where spatial, emission, and excitation dimensions jointly encode discriminative information. Our approach combines three components: a 3D convolutional autoencoder with parallel excitation branches for unsupervised representation learning, perturbation-based attribution for tracing latent dimension importance to individual excitation-emission wavelength pairs, and Maximum Marginal Relevance selection for balancing informativeness with spectral diversity.

Evaluated on lichen fluorescence imagery across 3,072 configurations, the framework demonstrates that strategically selected wavelengths can *exceed* baseline classification performance while reducing the data size —achieving 95.2% accuracy with 80 bands versus the 88.2% baseline with all 192 bands, and maintaining above-baseline accuracy down to 9 bands (95.3% data reduction). Robustness analysis confirms genuine selection intelligence (outperforming all 10,000 random combinations with 12.3σ separation), and object-level analysis shows that challenging specimens benefit most, with performance variance reduced by 54%. The 9-band optimal configuration selects one wavelength per excitation source, providing directly actionable sensor design guidelines.

To our knowledge, this work represents the first application of latent space perturbation analysis for wavelength selection in hyperspectral imaging, and the first deep learning framework specifically designed for 4D multi-excitation data. The approach is entirely unsupervised, requiring no class labels for wavelength selection, and produces ordered rankings that can be thresholded at any desired reduction level. The framework's modular design—separating representation learning, attribution, and selection—enables component substitution for adaptation to diverse spectroscopic applications.

Several directions merit future investigation:

- **Transfer learning:** Enabling wavelength selections learned on one sample type to generalize to related specimens, reducing the need for per-application autoencoder training.
- **Uncertainty quantification:** Incorporating Bayesian or variational autoencoders to provide confidence intervals on wavelength importance, valuable when selection guides expensive sensor design decisions.
- **End-to-end optimization:** Jointly training the autoencoder and selection mechanism for downstream tasks, potentially improving task-specific performance at the cost of requiring labeled data.
- **Multi-objective selection:** Simultaneously optimizing for multiple downstream tasks (e.g., species classification and stress detection) to address applications with diverse analytical requirements.
- **Active learning integration:** Creating closed-loop systems that iteratively refine wavelength selection based on acquisition results, converging toward optimal spectral sampling for specific samples.

As multi-excitation imaging systems become more prevalent across biomedical, agricultural, and environmental applications, principled approaches to spectral information extraction—like the framework presented here—will be essential for translating data richness into practical insight. The demonstrated ability to improve classification accuracy while reducing the data size opens new possibilities for real-time analysis, compact sensor design, and efficient data transmission in spectral imaging applications.

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